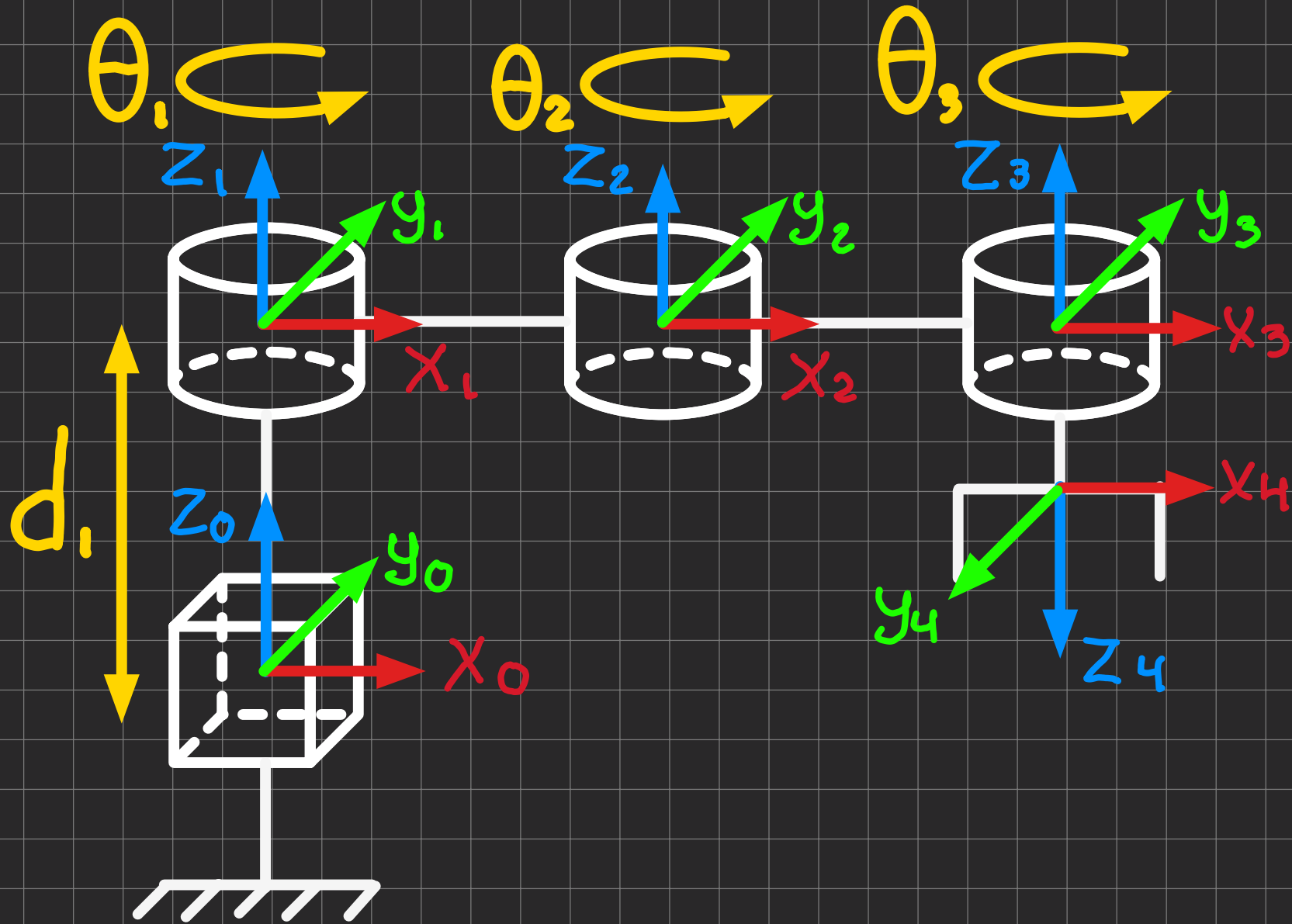


Cinemática Scara



Parámetros DH

	a	α	θ	d	
l_1	0	0	0	q_1	$\rightarrow q_1 = d_1$
l_2	a_2	0	q_2	0	$\rightarrow q_2 = \theta_1$
l_3	a_3	0	q_3	0	$\rightarrow q_3 = \theta_2$
l_4	0	π	q_4	$-d_4$	$\rightarrow q_4 = \theta_3$

Matrices T. Homogéneas

$$A_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} \cos q_2 & -\sin q_2 & 0 & a_2 \cos q_2 \\ \sin q_2 & \cos q_2 & 0 & a_2 \sin q_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} \cos q_3 & -\sin q_3 & 0 & a_3 \cos q_3 \\ \sin q_3 & \cos q_3 & 0 & a_3 \sin q_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} \cos q_4 & \sin q_4 & 0 & 0 \\ \sin q_4 & -\cos q_4 & 0 & 0 \\ 0 & 0 & -1 & -d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{12} = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & a_2 \cos \theta_2 \\ \sin \theta_2 & \cos \theta_2 & 0 & a_2 \sin \theta_2 \\ 0 & 0 & 1 & \theta_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{13} = \begin{bmatrix} \cos \theta_{23} & -\sin \theta_{23} & 0 & a_2 \cos \theta_2 + a_3 \cos \theta_{23} \\ \sin \theta_{23} & \cos \theta_{23} & 0 & a_2 \sin \theta_2 + a_3 \sin \theta_{23} \\ 0 & 0 & 1 & \theta_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{14} = \begin{bmatrix} \cos \theta_{24} & -\sin \theta_{24} & 0 & a_2 \cos \theta_2 + a_3 \cos \theta_{23} \\ \sin \theta_{24} & \cos \theta_{24} & 0 & a_2 \sin \theta_2 + a_3 \sin \theta_{23} \\ 0 & 0 & -1 & \theta_1 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\theta_{23} = \theta_2 + \theta_3$$

$$\theta_{24} = \theta_{23} + \theta_4$$